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Q1.a

For a SpMV node

Bc min $\geq 6 B/F$

```
do i = 1, Nr
  do j = row_ptr(i), row_ptr(i+1) - 1
    C(i) = C(i) + val(j) * B(col_idx(j))
  enddo
enddo
```

```
real*8    val(Nnz)
integer*4  col_idx(Nnz)
integer*4  row_ptr(Nr)
real*8    C(Nr)
real*8    B(Nc)
```

Min. load traffic [B]: $(8 + 4) N_{nz} + (4 + 8) N_r + 8 N_c$

Min. store traffic [B]: $8 N_r$

Total FLOP count [F]: $2 N_{nz}$

$$B_{c,min} = \frac{12 N_{nz} + 20 N_r + 8 N_c}{2 N_{nz}} \frac{B}{F} = \frac{12 + 20/N_{nzc} + 8/N_{nzc}}{2} \frac{B}{F}$$

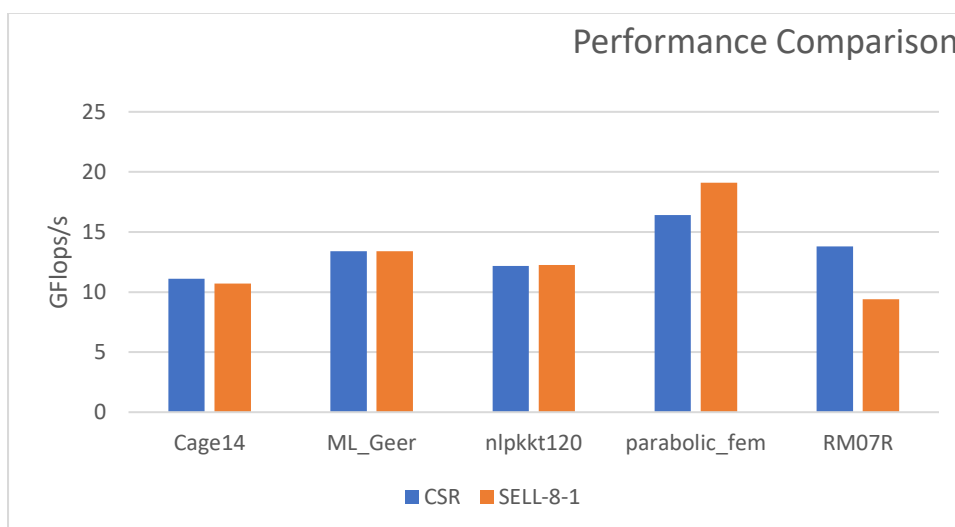
Nonzeros per row ($N_{nzc} = N_{nz}/N_r$) or column ($N_{nzc} = N_{nz}/N_c$)

Lower bound for code balance: $B_{c,min} \geq 6 \frac{B}{F} \rightarrow I_{max} \leq \frac{1}{6} \frac{F}{B}$

Roofline performance = $B_s/B_{cmin} = 85/6 = 14.166$

B)

	CSR (GFlops)	SELL-8-1 (GFLOPS)
Cage14	11.1	10.7
ML_Geer	13.4	13.4
nlpkkt120	12.16	12.24
parabolic_fem	16.4	19.1
RM07R	13.8	9.4

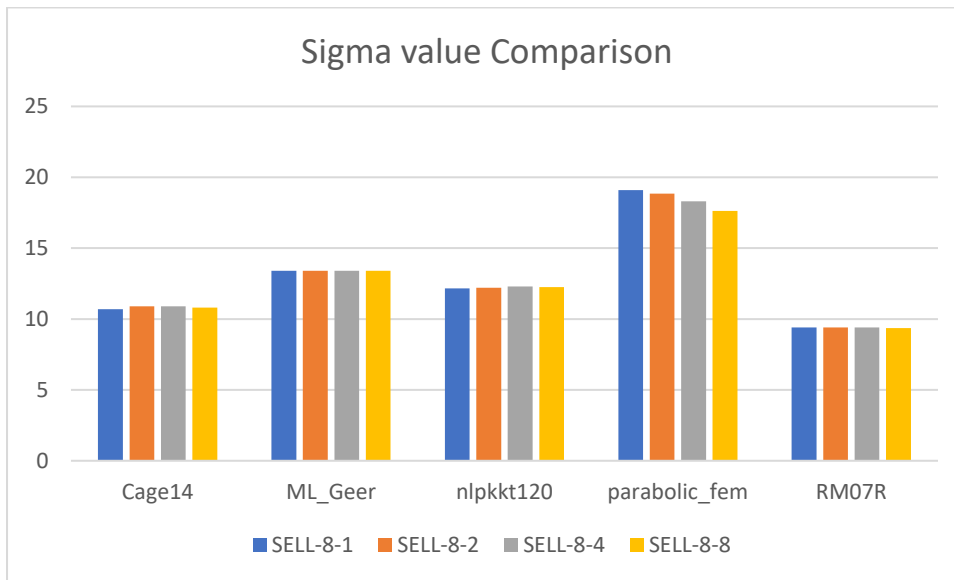


For parabolic_fem we are witnessing more performance than the matrix independent.(Screen shots of results attached)

c)

Unit = GFLOP/s

	SELL-8-1	SELL-8-2	SELL-8-4	SELL-8-8
Cage14	10.7	10.9	10.9	10.8
ML_Geer	13.4	13.4	13.4	13.4
nlpkkt120	12.16	12.21	12.29	12.25
parabolic_fem	19.1	18.85	18.3	17.62
RM07R	9.4	9.4	9.4	9.367



(sigma from 1 to 8)

In the SELL-C- σ storage format, the parameter sigma (σ) represents the sorting scope, which determines the maximum number of non-zero elements considered for sorting within each chunk. The sorting scope impacts the trade-off between memory consumption and computational efficiency.

A larger sorting scope (sigma > 1) allows for a wider range of non-zero elements to be considered for sorting within each chunk. If the matrix exhibits irregular or clustered non-zero patterns, a larger sorting scope can improve cache utilization

But in our case we are not witnessing any significant change with increasing sigma.

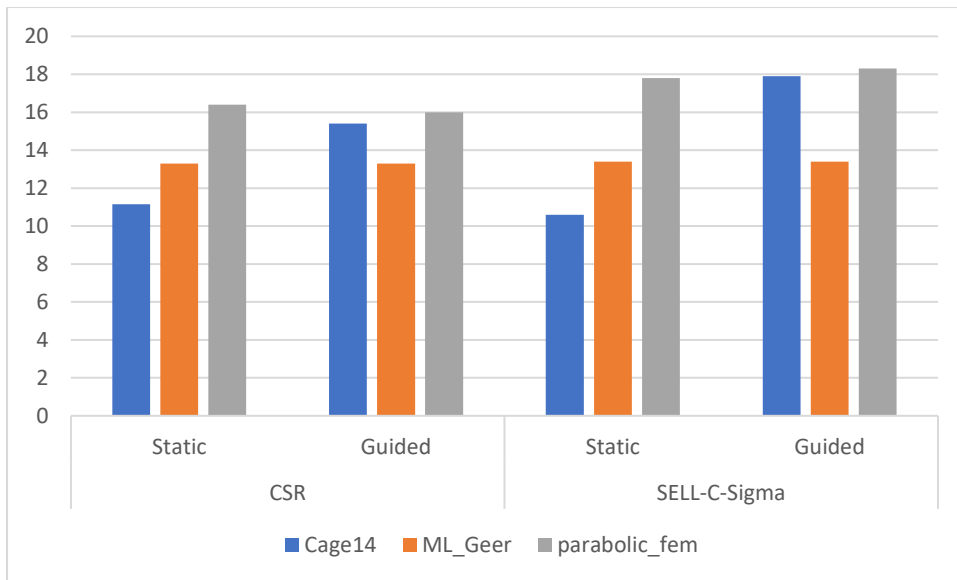
d)

After running the code using MEM_DP metric group for cage14, ML_Geer, and parabolic_fem We get the following performance values for static and dynamic omp scheduling

Performance of the matrices are as follows:

After running

	CSR		SELL-C-Sigma	
	Static	Guided	Static	Guided
Cage14	11.15	15.4	10.6	17.9
ML_Geer	13.3	13.3	13.4	13.4
parabolic_fem	16.4	16	17.8	18.3



We can see a performance upgrade only in Cage14, which makes sense if we look closer into the hardware performance stats.

Cage 14 Static SELL-C-SIGMA

We can see here the instruction retired is distributed un evenly as the max and min has a huge difference which means that the load was not properly distributed. Hence a non static scheduling method could help as Non-static scheduling allows for more dynamic load balancing and can better handle irregular workloads, which are common in sparse computations.

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	7872003400	338185800	656532000	4.373335e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	21657732000	1191238000	1209837000	1.203207e+09
CPU_CLK_UNHALTED_REF STAT	FIXC2	21657729000	1191231000	1209838000	1.203207e+09
TOPDOWN_SLOTS STAT	FIXC3	108288670000	5956191000	6049186000	6.016037e+09
PWR_PKG_ENERGY STAT	PWR0	79.3816	0	79.3816	4.4101
PWR_DRAM_ENERGY STAT	PWR3	7.2563	0	7.2563	0.4031
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	576	32	32	32
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	1394619600	43297200	106589600	7.747887e+07
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	144343300	0	144343300	8.019072e+06
CAS_COUNT_WR STAT	MBOX0C1	4420207	0	4420207	245567.0556
CAS_COUNT_RD STAT	MBOX1C0	144294600	0	144294600	8.016367e+06
CAS_COUNT_WR STAT	MBOX1C1	4406251	0	4406251	244791.7222
CAS_COUNT_RD STAT	MBOX2C0	144334600	0	144334600	8.018589e+06
CAS_COUNT_WR STAT	MBOX2C1	4405590	0	4405590	244755
CAS_COUNT_RD STAT	MBOX3C0	144352400	0	144352400	8.019578e+06
CAS_COUNT_WR STAT	MBOX3C1	4411319	0	4411319	245073.2778
CAS_COUNT_RD STAT	MBOX4C0	2204	0	2204	122.4444
CAS_COUNT_WR STAT	MBOX4C1	2147	0	2147	119.2778
CAS_COUNT_RD STAT	MBOX5C0	4127	0	4127	229.2778
CAS_COUNT_WR STAT	MBOX5C1	2814	0	2814	156.3333
CAS_COUNT_RD STAT	MBOX6C0	2686	0	2686	149.2222
CAS_COUNT_WR STAT	MBOX6C1	2734	0	2734	151.8889
CAS_COUNT_RD STAT	MBOX7C0	3806	0	3806	211.4444
CAS_COUNT_WR STAT	MBOX7C1	2540	0	2540	141.1111

Commands used :

OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 1

OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1

OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 1

OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 1

OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1

OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 1

(Further Screen shots attached.)

E) Lets look at stats for cage14

CSR

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	14652608700	768973400	984390500	8.140338e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	20788023000	1143078000	1159145000	1.154890e+09
CPU_CLK_UNHALTED_REF STAT	FIXC2	20788012000	1143073000	1159145000	1.154890e+09
TOPDOWN_SLOTS STAT	FIXC3	103940127000	5715390000	5795726000	5774451500
PWR_PKG_ENERGY STAT	PWR0	77.8534	0	77.8534	4.3252
PWR_DRAM_ENERGY STAT	PWR3	7.0937	0	7.0937	0.3941
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	148888900	7546300	8365500	8.271606e+06
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	1354113740	63647230	81745630	7.522854e+07
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	1241322500	38345900	95318600	6.896236e+07
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	142176800	0	142176800	7.898711e+06
CAS_COUNT_WR STAT	MBOX0C1	4413610	0	4413610	245200.5556
CAS_COUNT_RD STAT	MBOX1C0	142168000	0	142168000	7.898222e+06
CAS_COUNT_WR STAT	MBOX1C1	4409991	0	4409991	244999.5000
CAS_COUNT_RD STAT	MBOX2C0	142168500	0	142168500	7898250
CAS_COUNT_WR STAT	MBOX2C1	4402878	0	4402878	244604.3333
CAS_COUNT_RD STAT	MBOX3C0	142228200	0	142228200	7.901567e+06
CAS_COUNT_WR STAT	MBOX3C1	4411454	0	4411454	245080.7778
CAS_COUNT_RD STAT	MBOX4C0	1920	0	1920	106.6667
CAS_COUNT_WR STAT	MBOX4C1	1846	0	1846	102.5556
CAS_COUNT_RD STAT	MBOX5C0	3836	0	3836	213.1111
CAS_COUNT_WR STAT	MBOX5C1	2169	0	2169	120.5000
CAS_COUNT_RD STAT	MBOX6C0	2174	0	2174	120.7778
CAS_COUNT_WR STAT	MBOX6C1	2147	0	2147	119.2778
CAS_COUNT_RD STAT	MBOX7C0	3521	0	3521	195.6111
CAS_COUNT_WR STAT	MBOX7C1	1990	0	1990	110.5556

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	7872003400	338185800	656532000	4.373335e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	21657732000	1191238000	1209837000	1.203207e+09
CPU_CLK_UNHALTED_REF STAT	FIXC2	21657729000	1191231000	1209838000	1.203207e+09
TOPDOWN_SLOTS STAT	FIXC3	108288670000	5956191000	6049186000	6.016037e+09
PWR_PKG_ENERGY STAT	PWR0	79.3816	0	79.3816	4.4101
PWR_DRAM_ENERGY STAT	PWR3	7.2563	0	7.2563	0.4031
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	576	32	32	32
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	1394619600	43297200	106589600	7.747887e+07
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	144343300	0	144343300	8.019072e+06
CAS_COUNT_WR STAT	MBOX0C1	4420207	0	4420207	245567.0556
CAS_COUNT_RD STAT	MBOX1C0	144294600	0	144294600	8.016367e+06
CAS_COUNT_WR STAT	MBOX1C1	4406251	0	4406251	244791.7222
CAS_COUNT_RD STAT	MBOX2C0	144334600	0	144334600	8.018589e+06
CAS_COUNT_WR STAT	MBOX2C1	4405590	0	4405590	244755
CAS_COUNT_RD STAT	MBOX3C0	144352400	0	144352400	8.019578e+06
CAS_COUNT_WR STAT	MBOX3C1	4411319	0	4411319	245073.2778
CAS_COUNT_RD STAT	MBOX4C0	2204	0	2204	122.4444
CAS_COUNT_WR STAT	MBOX4C1	2147	0	2147	119.2778
CAS_COUNT_RD STAT	MBOX5C0	4127	0	4127	229.2778
CAS_COUNT_WR STAT	MBOX5C1	2814	0	2814	156.3333
CAS_COUNT_RD STAT	MBOX6C0	2686	0	2686	149.2222
CAS_COUNT_WR STAT	MBOX6C1	2734	0	2734	151.8889
CAS_COUNT_RD STAT	MBOX7C0	3806	0	3806	211.4444
CAS_COUNT_WR STAT	MBOX7C1	2540	0	2540	141.1111

The performance counter FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE represents the number of retired floating-point arithmetic instructions executed in packed mode for 256-bit packed double-precision data types. The performance counter FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE represents the number of retired floating-point arithmetic instructions executed in packed mode for 256-bit packed double-precision data types. In this case, the SIMD width is 256 bits, allowing for simultaneous processing of four double-precision floating-point values.

The average no of packed double-precision floating-point operations being executed by the processor for CSR and SELL-C-Sigma is 6.89×10^7 and 7.74×10^7 respectively So we can say that the two kernels are vectorized

F)

With SIMD

```
ptfs214h@f0907:~/spmvms_OMP_SCHEDULE=static$ run --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:      2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 1
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 118879972 (73.7233 per Row)
Starting CSR
runtime CSR: 8.87966 sec. performance: 2.49739e+09 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 8.75242 sec. performance: 2.5337e+09 Flops/s, overhead: 0.274206%
-----
Region SpMV_CSR, Group 1: MEM_DP
+-----+
| Region Info | HWThread 0 |
+-----+
| RDTSR Runtime [s] | 8.878868 |
| call count | 1 |
+-----+
-----
+-----+
| Event | Counter | HWThread 0 |
+-----+
| INSTR_RETIRED_ANY | FIXC0 | 25063340000 |
| CPU_CLK_UNHALTED_CORE | FIXC1 | 21286410000 |
| CPU_CLK_UNHALTED_REF | FIXC2 | 21286410000 |
| TOPDOWN_SLOTS | FIXC3 | 106032000000 |
| PWR_PKG_ENERGY | PWR0 | 950.6282 |
| PWR_DRAM_ENERGY | PWR3 | 79.0473 |
| FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE | PRC0 | 150400200 |
| FP_ARITH_INST_RETIRED_SCALAR_DOUBLE | PRC1 | 988778800 |
| FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE | PRC2 | 5502404000 |
| FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE | PRC3 | 0 |
| CAS_COUNT_RD | MBOX0C0 | 532534500 |
| CAS_COUNT_WR | MBOX0C1 | 4397960 |
| CAS_COUNT_RD | MBOX1C0 | 532495100 |
| CAS_COUNT_WR | MBOX1C1 | 4376428 |
| CAS_COUNT_RD | MBOX2C0 | 532493600 |
| CAS_COUNT_WR | MBOX2C1 | 4382652 |
| CAS_COUNT_RD | MBOX3C0 | 532569900 |
| CAS_COUNT_WR | MBOX3C1 | 4382636 |
| CAS_COUNT_RD | MBOX4C0 | 34738 |
| CAS_COUNT_WR | MBOX4C1 | 33427 |
+-----+
```

Without SIMD

```
ptfs214h@f0907:~/novect$ module load intel likwid
ptfs214h@f0907:~/novect$ vim Makefile
ptfs214h@f0907:~/novect$ make
icx -O3 -xHOST -no-vec -DVERBOSE -fopenmp -DUSE_LIKWID -I/apps/likwid/5.2.2/include -DLIKWID_PERFMON -c -o mmio/mmio.o mmio/mmio.c
icpx -O3 -xHOST -no-vec -DVERBOSE -fopenmp -DUSE_LIKWID -I/apps/likwid/5.2.2/include -DLIKWID_PERFMON -std=c++11 -c MMreader.cpp
icpx -O3 -xHOST -no-vec -DVERBOSE -fopenmp -DUSE_LIKWID -I/apps/likwid/5.2.2/include -DLIKWID_PERFMON -std=c++11 -c CSRMatrix.cpp
icx -O3 -xHOST -no-vec -DVERBOSE -fopenmp -DUSE_LIKWID -I/apps/likwid/5.2.2/include -DLIKWID_PERFMON -c -o timing/timing.o timing/timing.c
icpx -O3 -xHOST -no-vec -DVERBOSE -fopenmp -DUSE_LIKWID -I/apps/likwid/5.2.2/include -DLIKWID_PERFMON -std=c++11 -c benchmark.cpp
benchmark.cpp:59:13: warning: enumeration value 'omp_sched_monotonic' not handled in switch [-Wswitch]
    switch (schedType)
      ^
~~~~~
warning: loop not vectorized: the optimizer was unable to perform the requested transformation; the transformation might be disabled or specified as
part of an unsupported transformation ordering [-Wpass-failed=transform-warning]
warning: loop not vectorized: the optimizer was unable to perform the requested transformation; the transformation might be disabled or specified as
part of an unsupported transformation ordering [-Wpass-failed=transform-warning]
warning: loop not vectorized: the optimizer was unable to perform the requested transformation; the transformation might be disabled or specified as
part of an unsupported transformation ordering [-Wpass-failed=transform-warning]
4 warnings generated.
icpx -O3 -xHOST -no-vec -DVERBOSE -fopenmp -DUSE_LIKWID -I/apps/likwid/5.2.2/include -DLIKWID_PERFMON -std=c++11 -o benchmark mmio/mmio.o MMreader.
o CSRMatrix.o timing/timing.o benchmark.o -L/apps/likwid/5.2.2/lib -llikwid -lm
ptfs214h@f0907:~/novect$ OMP_SCHEDULE=static$ run --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:      2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 1
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 118879972 (73.7233 per Row)
Starting CSR
runtime CSR: 9.46296 sec. performance: 2.34345e+09 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 10.6965 sec. performance: 2.0732e+09 Flops/s, overhead: 0.274206%
-----
Region SpMV_CSR, Group 1: MEM_DP
+-----+
| Region Info | HWThread 0 |
+-----+
```

```

ptfs214h@f0907:~/novac$ OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/HL_Geer.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 1
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 9.46296 sec. performance: 2.34345e+09 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 10.6965 sec. performance: 2.0732e+09 Flops/s, overhead: 0.274206%
-----
Region SpMV_CSR, Group 1: MEM_DP
+-----+-----+-----+
| Region Info | HWThread 0 |
+-----+-----+-----+
| RDTSC Runtime [s] | 9.462174 |
| call count | 1 |
+-----+-----+-----+

+-----+-----+-----+
| Event | Counter | HWThread 0 |
+-----+-----+-----+
| INSTR_RETIRED_ANY | FIXC0 | 53958570000 |
| CPU_CLK_UNHALTED_CORE | FIXC1 | 22599140000 |
| CPU_CLK_UNHALTED_REF | FIXC2 | 22599130000 |
| TOPDOWN_SLOTS | FIXC3 | 11299570000 |
| PWR_PKG_ENERGY | PWR0 | 1012.7440 |
| PWR_DRAM_ENERGY | PWR3 | 81.9298 |
| FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE | PMC0 | 2697597000 |
| FP_ARITH_INST_RETIRED_SCALAR_DOUBLE | PMC1 | 5990411000 |
| FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE | PMC2 | 2697597000 |
| FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE | PMC3 | 0 |
| CAS_COUNT_RD | MBOX0C0 | 533113800 |
| CAS_COUNT_WR | MBOX0C1 | 4467277 |
| CAS_COUNT_RD | MBOX1C0 | 533072700 |
| CAS_COUNT_WR | MBOX1C1 | 4448877 |
+-----+-----+-----+

```

We don't see a lot of performance degradation with

In CSR it goes from 2.4 GFlop/s to 2.3 GFlop/s and in SELL-C-SIGM it goes from 2.5 GFlop/s to 2.0 GFlop/s.

But if we compare the FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE values:

CSR with SIMD

Event	Counter	HWThread 0
INSTR_RETIRED_ANY	FIXC0	25063340000
CPU_CLK_UNHALTED_CORE	FIXC1	21206410000
CPU_CLK_UNHALTED_REF	FIXC2	21206410000
TOPDOWN_SLOTS	FIXC3	10603200000
PWR_PKG_ENERGY	PWR0	950.6282
PWR_DRAM_ENERGY	PWR3	79.0473
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE	PMC0	150400200
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE	PMC1	908778800
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE	PMC2	5542404000
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE	PMC3	0
CAS_COUNT_RD	MBOX0C0	532534500
CAS_COUNT_WR	MBOX0C1	4397069
CAS_COUNT_RD	MBOX1C0	532495100
CAS_COUNT_WR	MBOX1C1	4376428
CAS_COUNT_RD	MBOX2C0	532493600
CAS_COUNT_WR	MBOX2C1	4382652
CAS_COUNT_RD	MBOX3C0	532560900
CAS_COUNT_WR	MBOX3C1	4382636
CAS_COUNT_RD	MBOX4C0	34730
CAS_COUNT_WR	MBOX4C1	33427
CAS_COUNT_RD	MBOX5C0	37543
CAS_COUNT_WR	MBOX5C1	38580
CAS_COUNT_RD	MBOX6C0	39088
CAS_COUNT_WR	MBOX6C1	38967
CAS_COUNT_RD	MBOX7C0	36083
CAS_COUNT_WR	MBOX7C1	35673

CSR without SIMD

Event	Counter	HWThread 0
INSTR_RETIRED_ANY	FIXC0	53958570000
CPU_CLK_UNHALTED_CORE	FIXC1	22599140000
CPU_CLK_UNHALTED_REF	FIXC2	22599130000
TOPDOWN_SLOTS	FIXC3	112995700000
PWR_PKG_ENERGY	PWR0	1012.7440
PWR_DRAM_ENERGY	PWR3	81.9298
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE	PMC0	2697597000
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE	PMC1	5990411000
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE	PMC2	2697597000
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE	PMC3	0
CAS_COUNT_RD	MBOX0C0	533113800
CAS_COUNT_WR	MBOX0C1	4467277
CAS_COUNT_RD	MBOX1C0	533072700
CAS_COUNT_WR	MBOX1C1	4448077
CAS_COUNT_RD	MBOX2C0	533074500
CAS_COUNT_WR	MBOX2C1	4448428
CAS_COUNT_RD	MBOX3C0	533098100
CAS_COUNT_WR	MBOX3C1	4449944
CAS_COUNT_RD	MBOX4C0	37748
CAS_COUNT_WR	MBOX4C1	36413
CAS_COUNT_RD	MBOX5C0	41160
CAS_COUNT_WR	MBOX5C1	42327
CAS_COUNT_RD	MBOX6C0	42826
CAS_COUNT_WR	MBOX6C1	42808
CAS_COUNT_RD	MBOX7C0	39309
CAS_COUNT_WR	MBOX7C1	38653

The FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE has decreased after adding the no-vec flag and SCALAR value has increased. Even more so is observed in SELL-C-SIGMA:

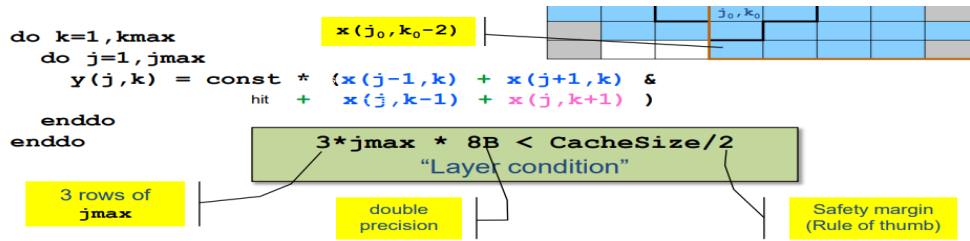
With SIMD

Event	Counter	HWThread 0
INSTR_RETIRED_ANY	FIXC0	15458770000
CPU_CLK_UNHALTED_CORE	FIXC1	20899520000
CPU_CLK_UNHALTED_REF	FIXC2	20899520000
TOPDOWN_SLOTS	FIXC3	104497600000
PWR_PKG_ENERGY	PWR0	936.9580
PWR_DRAM_ENERGY	PWR3	79.1300
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE	PMC0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE	PMC1	32
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE	PMC2	5559242000
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE	PMC3	0
CAS_COUNT_RD	MBOX0C0	534304700
CAS_COUNT_WR	MBOX0C1	4474640
CAS_COUNT_RD	MBOX1C0	534298500
CAS_COUNT_WR	MBOX1C1	4456598
CAS_COUNT_RD	MBOX2C0	534321500
CAS_COUNT_WR	MBOX2C1	4463121
CAS_COUNT_RD	MBOX3C0	534340400
CAS_COUNT_WR	MBOX3C1	4460501
CAS_COUNT_RD	MBOX4C0	34112
CAS_COUNT_WR	MBOX4C1	33124
CAS_COUNT_RD	MBOX5C0	36597
CAS_COUNT_WR	MBOX5C1	37746
CAS_COUNT_RD	MBOX6C0	38160
CAS_COUNT_WR	MBOX6C1	38242
CAS_COUNT_RD	MBOX7C0	35440
CAS_COUNT_WR	MBOX7C1	35221

Without SIMD

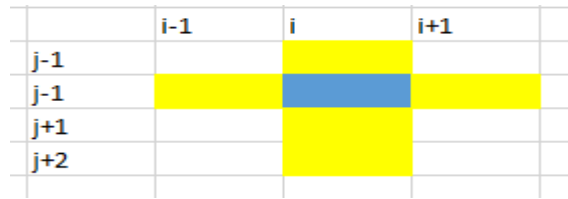
Event	Counter	HWThread 0
INSTR_RETIRED_ANY	FIXC0	42181930000
CPU_CLK_UNHALTED_CORE	FIXC1	25547090000
CPU_CLK_UNHALTED_REF	FIXC2	25547090000
TOPDOWN_SLOTS	FIXC3	127735400000
PWR_PKG_ENERGY	PWR0	1134.2340
PWR_DRAM_ENERGY	PWR3	92.9865
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE	PMC0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE	PMC1	21631740000
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE	PMC2	151306400
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE	PMC3	0
CAS_COUNT_RD	MBOX0C0	532027900
CAS_COUNT_WR	MBOX0C1	4448764
CAS_COUNT_RD	MBOX1C0	532041900
CAS_COUNT_WR	MBOX1C1	4434505
CAS_COUNT_RD	MBOX2C0	532093400
CAS_COUNT_WR	MBOX2C1	4432846
CAS_COUNT_RD	MBOX3C0	532146000
CAS_COUNT_WR	MBOX3C1	4437882
CAS_COUNT_RD	MBOX4C0	42447
CAS_COUNT_WR	MBOX4C1	41253
CAS_COUNT_RD	MBOX5C0	45681
CAS_COUNT_WR	MBOX5C1	47149
CAS_COUNT_RD	MBOX6C0	48230
CAS_COUNT_WR	MBOX6C1	48237
CAS_COUNT_RD	MBOX7C0	44118
CAS_COUNT_WR	MBOX7C1	43482

Q2)



- (a) The relevant layer condition without assuming parallelization would be
 No. of rows to be fitted in cache to calculate $y=4$
 Double precision and $j_{\max}=M$
 Safety margin=2

$$4 * M * 8 \text{ Bytes} < \text{Cache size}/2$$



But since there is a parallelization of outer loop

$$nthreads * 4 * M * 8 \text{ Bytes} < \text{Cache size}/2$$

- (b) Worst case will require Load and store (including write allocate), 4 Load ups for $X + 1$ Load up for f . While in best case if the 4 rows are in the cache already then $x[j+2][i]$, $x[j+1][i]$, $x[j][i+1]$ will come from L2 or L3 cache and only $x[j-1][i]$, f and y need to be loaded

$$\text{Best code balance} = 8 + 8 + 16 = 32 \frac{B}{LUP}$$

$$\text{Worst code balance} = 8 * 4 + 8 + 16 = 56 \frac{B}{LUP}$$

- (c) Assuming in-memory problem with data not fitting into caches, the worst code balance is used.

$$\text{Absolute upper performance limit} = \frac{bs}{Bc} = \frac{150}{56} = 2.67 \text{ GLUP/s}$$

L1 and L2 are local caches. Each Fritz socket has 36 cores. Layer condition is independent of threads since L1 and L2 have Cache size=threads*Cache size

L1 cache:

$$nthreads * 4 * M * 8 \text{ Bytes} < \frac{48 * 1024}{2} * nthreads$$

$$32M < 24576$$

$$M < 768$$

L2 cache:

$$nthreads * 4 * M * 8 \text{ Bytes} < \frac{1.25 * 1024 * 1024}{2} * nthreads$$

$$32M < 23592960$$

$$M < 20480$$

L3 cache is a shared cache and an exclusive victim cache:

$$nthreads * 4 * M * 8 \text{ Bytes} < \frac{(54 + 1.25 + 48/1024) * 1024 * 1024}{2}$$

$$nthreads * 4 * M * 8 \text{ Bytes} < \frac{55.29 * 1024 * 1024}{2}$$

Assume $nthreads=36$ for Fritz socket

$$M < 25163$$